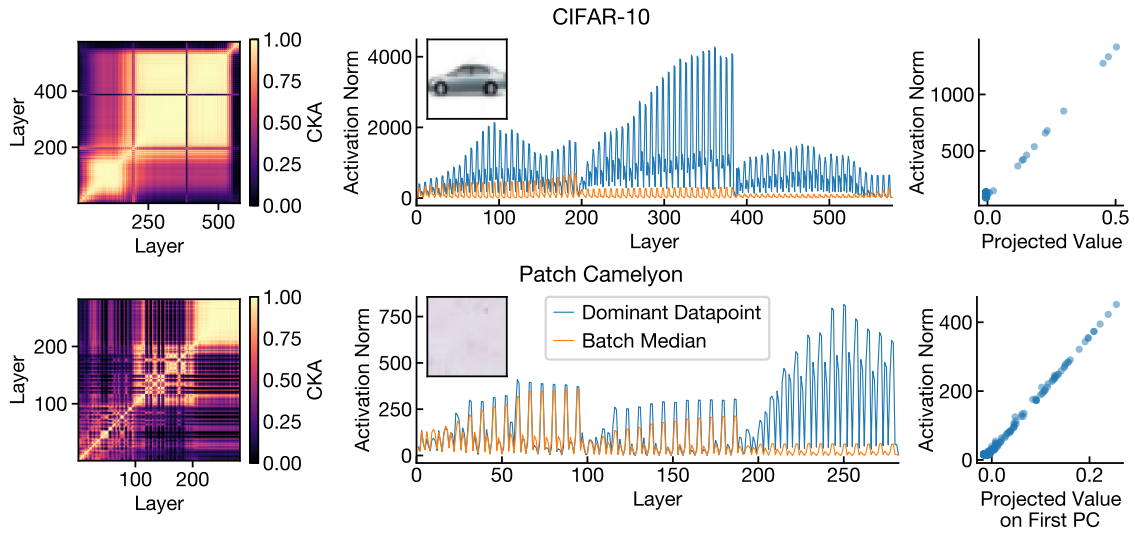

Figure 5: Datapoints that dominate the first principal components of the block structure also strongly activate the corresponding layers. We explore the relationship between dominant datapoints and activation norms for ResNet-164 (1×) on CIFAR-10 (top row) and ResNet-80 (1×) on Patch Camelyon (bottom row). For layers inside the block structure (left column), dominant datapoints (inset) produce much larger activation norms than the median norm of a random minibatch (middle column). Moreover, across datapoints, activation norms are highly correlated with their projections on the first principal component (right column).

the model. The right column of Figure 3 shows the corresponding properties of an architecturally identical model trained from a different seed, where the dominant datapoints share white backgrounds instead. Refer to Appendix D for visualizations of dominant images found in other tasks (CIFAR-100, Patch Camelyon) and model architectures (wide ResNet). Besides background color, the dominant datapoints can also reflect other simple image patterns that are prevalent in the dataset, such as the appearance of large dark spots in histopathologic scans (Appendix Figure D.3).

Finally, the block structure observed in linear CKA heatmaps arises solely from dominant datapoints. As seen in Figure 4, when the 10% most dominant datapoints are excluded from evaluation, the block structure is completely eliminated in all 3 training runs of ResNet-164 (1×) that we examined, and the fraction of variance explained by the first PCs is substantially reduced. In fact, for one training run, removing only the 1% most dominant datapoints is sufficient to achieve this effect. Thus, the block structure is completely determined by the dominant images, and is sensitive to the frequency of the dataset statistics that it captures.

4.2 Dominant Datapoints Have Large Activation Norms

We investigate what happens to the activations of a dominant example as it propagates through the network, and observe that it strongly activates the parts of the network with a visible block structure. Figure 5 shows two dominant datapoints, for a ResNet trained on CIFAR-10 that has a preference for white background images (top row), and for another ResNet trained on Patch Camelyon that responds to clear pink backgrounds (bottom row). Both models contain block structure in their internal representations, and we find that in the corresponding layers, the activations of the dominant datapoints are substantially larger in norm than the median activations of the minibatches they are

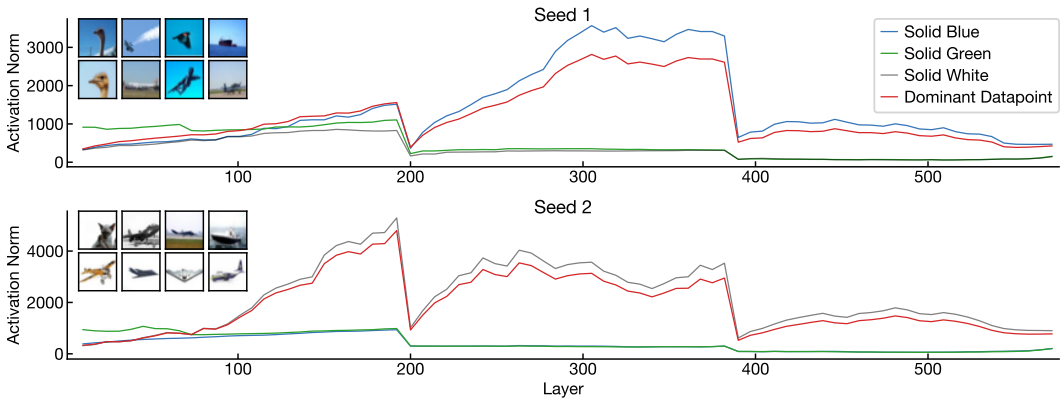

Figure 6: Solid color images strongly activate intermediate layers. Rows show ResNet-164 (1×) models that are trained from different random initializations. The top model's dominant datapoints consist of images with blue backgrounds (see inset images), whereas the bottom model prefers white background images. Layers of the top model are strongly activated by solid blue images, but not solid white images, whereas the bottom model shows the opposite pattern. Layers of both models are strongly activated by their respective dominant datapoints (red lines), but other solid colors (e.g., green) do not yield strong activations in either. To improve readability of the plot, we plot only the representations at the end of each ResNet block. See Appendix E for similar findings on Patch Camelyon models.

a part of. Moreover, the magnitude of the projection on the first PC is correlated with the activation norm. We conclude that dominant datapoints evoke activations with large norms, and activations of different dominant datapoints point in similar directions.

In Appendix J, we include additional results measuring representational similarity in networks with block structure using CKA with different kernels. All kernels we have tested are sensitive to differences between dominant and non-dominant datapoints, but the linear kernel produces particularly strong "blocks" compared to RBF or cosine kernels. Across all kernels, the removal of dominant datapoints consistently eliminates blocks in representational similarity heatmaps.

4.3 Case Study: Image Backgrounds As a Dominant Dataset Property

As background colors appear to be a common dataset characteristic that is picked up by many large-capacity models, we provide further evidence, through data and training manipulations, to confirm that this is a real property of the hidden representations that emerges only with overparameterization.

First, we attempt to illustrate the connection between specific background colors, which vary across random seeds, and the layer activations. To approximate this data statistic, we repeat only the top left pixel in each image across the entire dimensions of the image, obtaining solid color images. These synthetic images indeed yield even larger activations compared to the dominant images they are taken from, and different initializations of architecturally identical networks respond to different synthetic images. For instance, given the ResNet-164 (1×) model that has dominant datapoints containing a blue background (Figure 3), its hidden layers are further activated when all image pixels are replaced with the same shade of blue, but a solid white image produces considerably smaller activations (Figure 6). In the same figure, we observe the opposite trend for another ResNet-164 (1×) seed, which has been shown to pick up on white backgrounds (see Figure 3). Refer to Appendix E for similar analysis on the Patch Camelyon dataset.

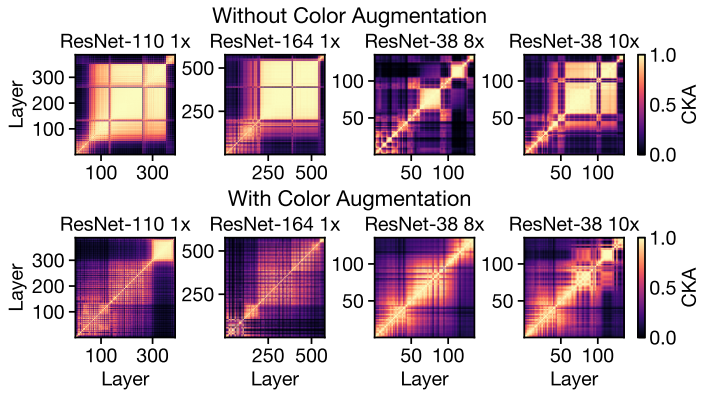

Figure 7: Simply adding color augmentation helps reduce the block structure effect. Having established that the activations and representational components of some large-capacity models pick up on common background colors from the inputs, we experiment with color dropping and color jittering during training to counter this effect. Indeed, color augmentation minimizes the block structure appearance in the internal representations.

Intervention: Color Augmentation: Given earlier insights, a natural intervention to prevent the network from potentially picking up background color signal is adding color augmentation. This includes randomly dropping color channels and jittering brightness, contrast, saturation, and hue of training images [16, 35]. As shown in Figure 7, training with this data augmentation reduces the block structure in large capacity models.

5 Evolution of Block Structure during Training

In the previous section, we characterize the signals the block structure propagates across its layers, and explore their implications on other aspects of the network internals. Informed by these findings, we next explore what happens to the block structure and the dominant images over time, from initialization until the model converges, and how this process varies across different training runs.

Figure 8 shows the evolution of the internal representations of a ResNet-110 (1×) model as it is trained for 300 epochs on CIFAR-10, and tracks how similar each checkpoint is to the final model. We observe that some structure in the CKA heatmap is already present by the first epoch, and the heatmap undergoes little qualitative change past epoch 20 (top set of plots). However, when we inspect the corresponding dominant datapoints and compare the hidden representations between intermediate checkpoints and the final model (bottom rows of plots), we find that the block structure does not always carry the same information. Instead, the representations, and corresponding groups of dominant datapoints, only stabilize much later in training. We observe similar behavior for other models in Appendix F, and note that the differences in block structure representations across random seeds already take shape near the start of training as well. Overall these findings suggest that the uniqueness of the block structure representations in large-capacity models can be attributed to both initialization parameters and the image minibatches the models receive throughout training.

To further explore the link between dominant datapoints and fluctuations in network representations, in Appendix Figure F.4, we track the magnitude of the projected value on the first PC of a single dominant datapoint found at the *end* of training, and find that the value plummets at epochs when the internal representation structure diverges from that of the fully trained model. At these epochs, the dominant datapoint does not produce large activations either (bottom set of plots). This illustrates that the precise set of dominant datapoints can vary significantly over the course of training for large-capacity models.